

GOVT 6019: Distribution Functions

Fall 2026

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1. Random variables and their distributions

There are two main types of random variables used in practice: **discrete** and **continuous**.

Definition (discrete random variable). A random variable X is *discrete* if there is a finite list of values $a_1, a_2, \dots, a_n$, or an infinite list of values $a_1, a_2, \dots$, such that $\mathbb{P}(X = a_j \text{ for some } j) = 1$. The finite or countably infinite set of values x such that $\mathbb{P}(X = x) > 0$ is called the **support** of X .¹

Most commonly in applications, the support of a discrete random variable is a set of integers. A **continuous** random variable, by contrast, can take on any real value in an interval, possibly the entire real line.

The **distribution** of a random variable specifies the probabilities of all events associated with it: the probability of it equaling 3, the probability of it being at least 110, and so on. There are several equivalent ways to express a distribution. For a discrete random variable the most natural is the probability mass function.

A note on notation. This handout writes f for both the probability mass function and the probability density function, and F for the cumulative distribution function, matching the lecture slides. Blitzstein and Hwang write p_X for the mass function and reserve f for the density.

2. The probability mass function

Definition (probability mass function). The *probability mass function* (PMF) of a discrete random variable X is the function f given by

$$f(x) = \mathbb{P}(X = x).$$

This is positive if x is in the support of X , and 0 otherwise.

In writing $\mathbb{P}(X = x)$ we are using $X = x$ to denote an **event**, consisting of all outcomes s to which X assigns the number x . This event is also written as $\{X = x\}$; formally, $\{X = x\}$ is defined as $\{s \in \Omega : X(s) = x\}$.²

¹The support is the set of values that X can actually come out as — formally, the set of x with $\mathbb{P}(X = x) > 0$.

²The colon reads “such that”: the set of all outcomes s in the sample space Ω such that X assigns s the number x . Gathering those outcomes produces a subset of Ω — an event — so taking its probability is defined. X itself is a function rather than a subset, which is why $\mathbb{P}(X)$ is not.

If X is the number of heads in two fair coin tosses, then $\{X = 1\}$ consists of the sample outcomes HT and TH , which are the two outcomes to which X assigns the number 1. Since $\{HT, TH\}$ is a subset of the sample space, it is an event. So it makes sense to talk about $\mathbb{P}(X = 1)$, or more generally $\mathbb{P}(X = x)$. If $\{X = x\}$ were anything other than an event, it would make no sense to calculate its probability.

It does **not** make sense to write $\mathbb{P}(X)$. We can only take the probability of an event, not of a random variable.

2.1 Example: two coin tosses

Consider two tosses of a fair coin, with sample space $\Omega = \{HH, HT, TH, TT\}$.

Let X be the number of heads. Since X equals 0 if TT occurs, 1 if HT or TH occurs, and 2 if HH occurs, the PMF of X is

$$f(0) = \frac{1}{4}, \quad f(1) = \frac{1}{2}, \quad f(2) = \frac{1}{4},$$

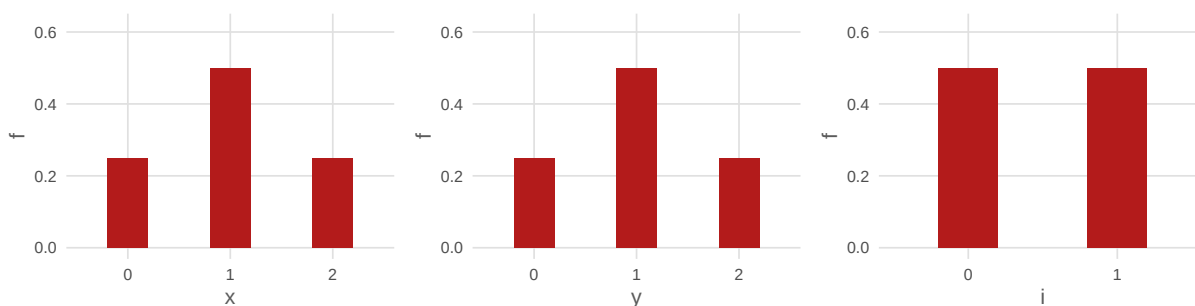
and $f(x) = 0$ for all other values of x .

Let $Y = 2 - X$, the number of tails. Reasoning as above, or using the fact that $\mathbb{P}(Y = y) = \mathbb{P}(2 - X = y) = \mathbb{P}(X = 2 - y)$, the PMF of Y is the same as the PMF of X .

Note that X and Y have the same PMF even though X and Y are not the same random variable: they are two different functions from $\{HH, HT, TH, TT\}$ to the real line.

Let I be the indicator of the first toss landing heads. Since I equals 0 if TH or TT occurs and 1 if HH or HT occurs,

$$f_I(0) = \frac{1}{2}, \quad f_I(1) = \frac{1}{2}.$$



2.2 Which functions are PMFs

Theorem (valid PMFs). Let X be a discrete random variable with support $x_1, x_2, \dots$. The PMF f of X must satisfy the following two criteria.

- **Nonnegative:** $f(x) > 0$ if $x = x_j$ for some j , and $f(x) = 0$ otherwise.
- **Sums to 1:** $\sum_{j=1}^{\infty} f(x_j) = 1$.

The first criterion is true since probability is nonnegative. The second is true since X must take on *some* value, and the events $\{X = x_j\}$ are disjoint, so

$$\sum_{j=1}^{\infty} \mathbb{P}(X = x_j) = \mathbb{P}\left(\bigcup_{j=1}^{\infty} \{X = x_j\}\right) = \mathbb{P}(X = x_1 \text{ or } X = x_2 \text{ or } \dots) = 1.$$

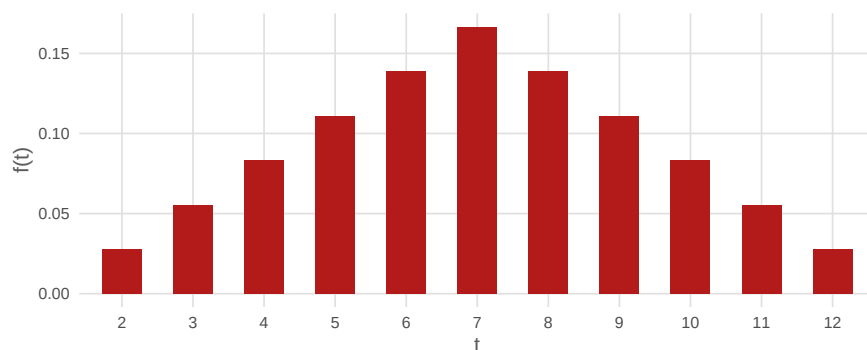
Conversely, if distinct values $x_1, x_2, \dots$ are specified and we have a function satisfying the two criteria above, then that function *is* the PMF of some random variable.

Two consequences follow immediately. Since the values are nonnegative and sum to 1, no single value can exceed 1: every $f(x)$ lies in $[0, 1]$. And since the events $\{X = x\}$ are disjoint, the probability that X falls in any set A of values is obtained by summing:

$$\mathbb{P}(X \in A) = \sum_{a \in A} f(a).$$

2.3 Example: the sum of two die rolls

Let T be the sum of two fair die rolls. Its support is $\{2, 3, \dots, 12\}$, and its PMF is plotted below. The distribution has a triangular shape, and it is symmetric: $\mathbb{P}(T = t) = \mathbb{P}(T = 14 - t)$.



Suppose we want the probability that T lies in the interval $[1, 4]$. There are only three values in $[1, 4]$ that T can take on, namely 2, 3 and 4. We know the probability of each of these from the PMF, so

$$\mathbb{P}(1 \leq T \leq 4) = \mathbb{P}(T = 2) + \mathbb{P}(T = 3) + \mathbb{P}(T = 4) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{6}{36}.$$

In general, given a discrete random variable X and a set B of real numbers, if we know the PMF of X we can find $\mathbb{P}(X \in B)$ by summing up the heights of the vertical bars at points in B . **Knowing the PMF of a discrete random variable determines its distribution.**

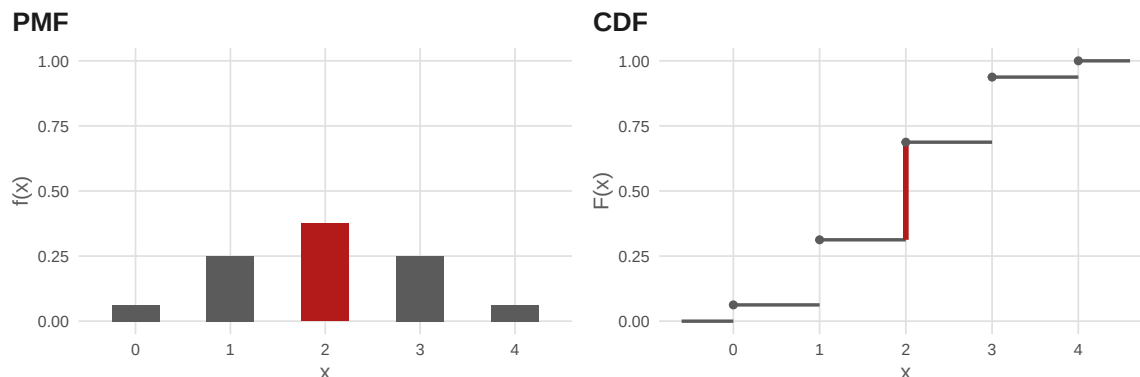
3. The cumulative distribution function

Definition (cumulative distribution function). The *cumulative distribution function* (CDF) of a random variable X is the function F given by $F(x) = \mathbb{P}(X \leq x)$.

Unlike the PMF, the CDF is defined for continuous random variables as well as discrete ones. For discrete random variables we can freely convert between CDF and PMF.

3.1 Example: converting between PMF and CDF

Let $X \sim \text{Binomial}(4, 1/2)$. The figure shows the PMF and CDF of X side by side. The height of the vertical bar at $x = 2$ in the PMF is also the height of the jump in the CDF at 2.



From PMF to CDF. To find $\mathbb{P}(X \leq 1.5)$, which is the CDF evaluated at 1.5, we sum the PMF over all values of the support that are less than or equal to 1.5:

$$\mathbb{P}(X \leq 1.5) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) = \left(\frac{1}{2}\right)^4 + 4\left(\frac{1}{2}\right)^4 = \frac{5}{16}.$$

Similarly, the value of the CDF at an arbitrary point x is the sum of the heights of the vertical bars of the PMF at values less than or equal to x .

From CDF to PMF. The CDF of a discrete random variable consists of jumps and flat regions. The height of a jump in the CDF at x is equal to the value of the PMF at x . In the figure, the height of the jump in the CDF at 2 is the same as the height of the corresponding vertical bar in the PMF:

$$f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16}.$$

The flat regions of the CDF correspond to values outside the support of X , so the PMF is equal to 0 in those regions.

3.2 Which functions are CDFs

Theorem (valid CDFs). Any CDF F has the following properties.

- **Increasing:** if $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$.
- **Right-continuous:** the CDF is continuous except possibly for having some jumps. Wherever there is a jump, the CDF is continuous from the right. That is, for any a , $F(a) = \lim_{x \rightarrow a^+} F(x)$.
- **Convergence to 0 and 1 in the limits:** $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

The first property holds because the event $\{X \leq x_1\}$ is a subset of the event $\{X \leq x_2\}$, so $\mathbb{P}(X \leq x_1) \leq \mathbb{P}(X \leq x_2)$. The third holds because X must take some value, so accumulating probability across the whole support gives 1.

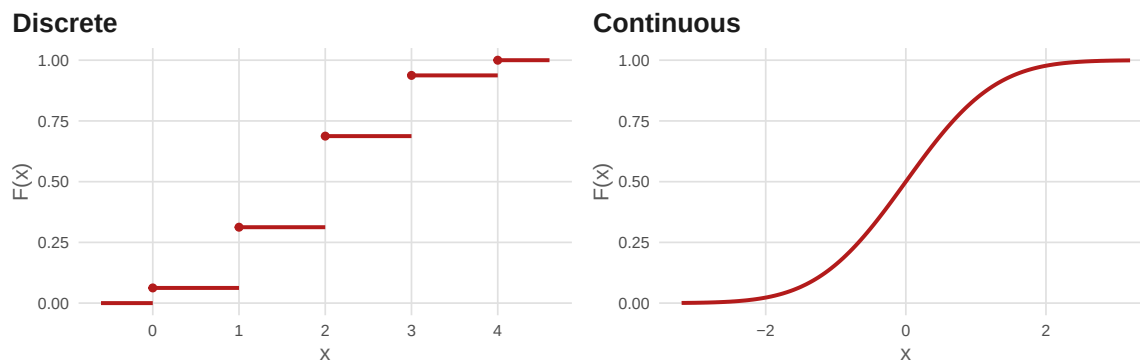
In the figure above, the CDF is increasing with some flat regions, continuous from the right, and converges to 0 and to 1. In this example it reaches 0 and 1; in some examples one or both of these values may be approached but never reached.

The converse is true as well: given any function F meeting these criteria, a random variable can be constructed whose CDF is F .

We have now seen two equivalent ways of expressing the distribution of a discrete random variable. The PMF and the CDF contain the same information, since we can always find the CDF from the PMF and the PMF from the CDF. Generally the PMF is easier to work with for discrete random variables, since evaluating the CDF requires a summation.

4. Continuous random variables and the density

Recall that for a discrete random variable the CDF jumps at every point in the support, and is flat everywhere else. For a continuous random variable the CDF increases smoothly.



Definition (continuous random variable). A random variable has a *continuous distribution* if its CDF is differentiable. A *continuous random variable* is a random variable with a continuous distribution.

For discrete random variables the CDF is awkward to work with because of its jumpiness, and its derivative is almost useless since it is undefined at the jumps and 0 everywhere else. For continuous random variables the CDF is often convenient to work with, and its derivative is a very useful function.

Definition (probability density function). For a continuous random variable X with CDF F , the *probability density function* (PDF) of X is the derivative f of the CDF, given by

$$f(x) = F'(x).$$

The *support* of X is the set of all x where $f(x) > 0$.

4.1 Why every single value has probability zero

An important way in which continuous random variables differ from discrete ones is that for a continuous random variable X ,

$$\mathbb{P}(X = x) = 0 \quad \text{for all } x.$$

This is because $\mathbb{P}(X = x)$ is the height of a jump in the CDF at x , and the CDF of a continuous random variable has no jumps. Since the PMF of a continuous random variable would be 0 everywhere, we work with a PDF instead.

Probability zero is not the same as impossible. Every individual value has probability zero, and yet the variable takes one of them on every draw.

4.2 Density is not probability

The PDF is analogous to the PMF in many ways, but there is a key difference. For a PDF f , the quantity $f(x)$ is **not** a probability, and in fact it is possible to have $f(x) > 1$ for some values of x . To obtain a probability we need to integrate the PDF.³

A valid PDF must satisfy two conditions, matching the two for a PMF with the sum replaced by an integral:

$$f(x) \geq 0 \text{ for all } x, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

It is the total **area** under the density that is constrained to 1, never the height.

4.3 From density to probability

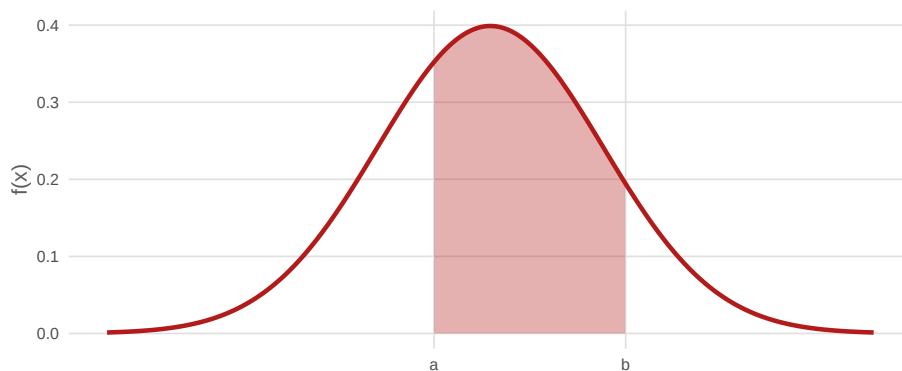
Proposition (PDF to CDF). Let X be a continuous random variable with PDF f . Then the CDF of X is given by

$$F(x) = \int_{-\infty}^x f(t) dt.$$

This is analogous to how we obtained the value of a discrete CDF at x by summing the PMF over all values less than or equal to x . Here we integrate the PDF over all values up to x , so **the CDF is the accumulated area under the PDF**. Since we can freely convert between the PDF and the CDF using the inverse operations of integration and differentiation, both the PDF and the CDF carry complete information about the distribution.

The probability that X falls in an interval is therefore an area, and it can equally be computed as a difference of two CDF values:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a).$$



The two readings of this formula are worth keeping separate. $F(b)$ is all the probability at or below b . $F(a)$ is all the probability at or below a . Everything counted in $F(a)$ is also counted in $F(b)$, so subtracting leaves exactly what lies between a and b .

³Another way to see this is with the following example. Take vote share. Measured as a proportion on $[0, 1]$, a density might be around 4 near the middle. Measure the identical variable as a percentage on $[0, 100]$ and every density value is divided by 100, because the horizontal axis got stretched by 100 while the total area stayed at 1. Same variable, same probabilities, density values a hundredfold apart. A quantity whose numerical value changes when you switch from proportions to percentages cannot be a probability. Probabilities are unitless.

Because the endpoints have probability 0, we can include or exclude them as we wish without altering the probability:

$$\mathbb{P}(a < X < b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a \leq X \leq b).$$

5. The Uniform distribution

Intuitively, a Uniform random variable on the interval (a, b) is a completely random number between a and b . We formalize the notion of “completely random” on an interval by specifying that the PDF should be **constant** over the interval.

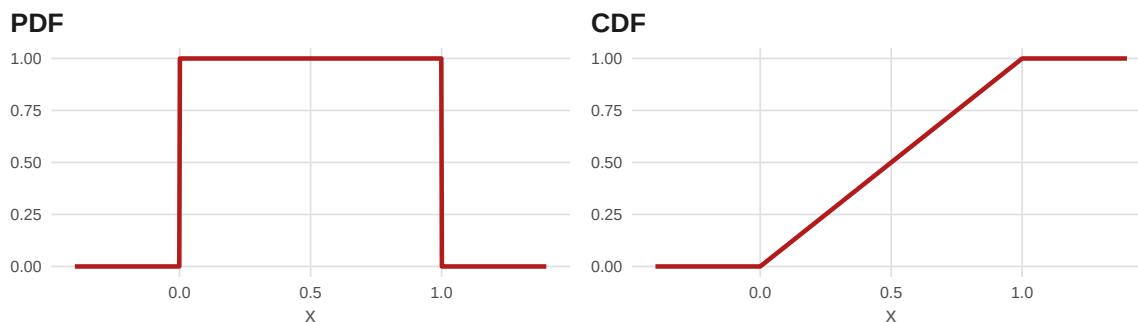
Definition (Uniform distribution). A continuous random variable X has the *Uniform distribution* on the interval (a, b) if its PDF is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

We denote this by $X \sim \mathcal{U}(a, b)$.

This is a valid PDF because the area under the curve is just the area of a rectangle with width $b - a$ and height $1/(b - a)$. The CDF is the accumulated area under the PDF:

$$F(x) = \begin{cases} 0 & \text{if } x \leq a, \\ \frac{x-a}{b-a} & \text{if } a < x < b, \\ 1 & \text{if } x \geq b. \end{cases}$$



The Uniform distribution used most frequently is $\mathcal{U}(0, 1)$, the **standard Uniform**, whose PDF and CDF are particularly simple: $f(x) = 1$ and $F(x) = x$ for $0 < x < 1$. For a general $\mathcal{U}(a, b)$ the PDF is constant on (a, b) , and the CDF is ramp-shaped, increasing linearly from 0 to 1 as x ranges from a to b .

For Uniform distributions, **probability is proportional to length**. If (c, d) is a subinterval of (a, b) of length $\ell = d - c$, then the probability of X being in (c, d) is proportional to ℓ : a subinterval twice as long has twice the probability of containing X , and a subinterval of the same length has the same probability. The constant of proportionality is $1/(b - a)$, so the probability is $\ell/(b - a)$.⁴

⁴Two quantities are proportional when one is a fixed multiple of the other: $y = kx$, with k the same for every x . That k is the constant of proportionality. Here it must be $1/(b - a)$, since the whole interval has length $b - a$ and probability 1.

6. The Normal distribution

We start with the simplest Normal distribution, the **standard Normal**, which is centered at 0 and has variance 1. From its properties we get to any Normal distribution by shifting and scaling.

Definition (standard Normal distribution). A continuous random variable Z has the *standard Normal distribution* if its PDF φ is given by

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

We write this as $Z \sim \mathcal{N}(0, 1)$, since Z has mean 0 and variance 1.

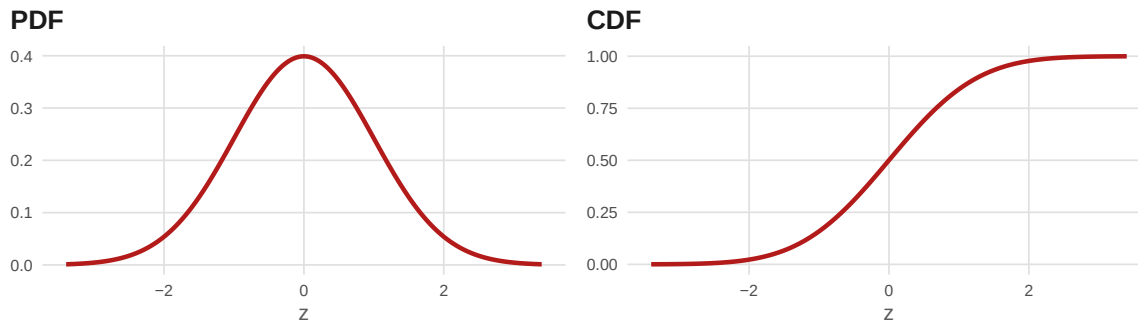
The constant $1/\sqrt{2\pi}$ in front of the PDF may look surprising. It is exactly what is needed to make the PDF integrate to 1. Such constants are called **normalizing constants**, because they normalize the total area under the PDF to 1. It sets the height of the curve, not its shape.

The standard Normal CDF Φ is the accumulated area under the PDF:

$$\Phi(z) = \int_{-\infty}^z \varphi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

It is left in terms of an integral because it is mathematically impossible to find a closed-form expression for the antiderivative of φ . Closed form or no, Φ is still a well-defined function: give it an input z , and it returns the accumulated area under the PDF from $-\infty$ up to z . In R, `pnorm()` evaluates Φ and `qnorm()` inverts it.

Notation. By convention we use φ for the standard Normal PDF and Φ for its CDF, and Z for a standard Normal random variable.



The PDF is bell-shaped and symmetric about 0. The CDF is *S*-shaped.

6.1 Symmetry

Three symmetry properties can be deduced from the standard Normal PDF and CDF.

1. **Symmetry of the PDF.** φ satisfies $\varphi(z) = \varphi(-z)$.
2. **Symmetry of tail areas.** The area under the PDF to the left of $-z$, which is $\mathbb{P}(Z \leq -z) = \Phi(-z)$, equals the area to the right of z , which is $\mathbb{P}(Z \geq z) = 1 - \Phi(z)$. In general,

$$\Phi(z) = 1 - \Phi(-z) \quad \text{for all } z.$$

3. **Symmetry of Z and $-Z$.** If $Z \sim \mathcal{N}(0, 1)$, then $-Z \sim \mathcal{N}(0, 1)$ as well.

6.2 The general Normal

The general Normal distribution has two parameters, μ and σ^2 , which correspond to the mean and variance. The standard Normal is the special case $\mu = 0$, $\sigma^2 = 1$.

Definition (Normal distribution). If $Z \sim \mathcal{N}(0, 1)$, then

$$X = \mu + \sigma Z$$

has the *Normal distribution* with mean μ and variance σ^2 , for any real μ and σ^2 with $\sigma > 0$. We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

That X has mean μ and variance σ^2 follows from the properties of expectation and variance:

$$\mathbb{E}(\mu + \sigma Z) = \mathbb{E}(\mu) + \sigma \mathbb{E}(Z) = \mu, \quad \mathbb{V}(\mu + \sigma Z) = \mathbb{V}(\sigma Z) = \sigma^2 \mathbb{V}(Z) = \sigma^2.$$

Note that we multiply Z by the standard deviation σ , not by σ^2 . Otherwise the units would be wrong and X would have variance σ^4 .

6.3 Standardization

If we can get from Z to X , then we can get from X back to Z . The process of getting a standard Normal from a non-standard Normal is called **standardization**. For $X \sim \mathcal{N}(\mu, \sigma^2)$, the standardized version of X is

$$\frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Standardization lets us write the CDF and PDF of any Normal in terms of the standard Normal CDF and PDF.

Theorem (Normal CDF and PDF). Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then the CDF of X is

$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right),$$

and the PDF of X is

$$f(x) = \varphi\left(\frac{x - \mu}{\sigma}\right) \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

For the CDF, start from the definition $F(x) = \mathbb{P}(X \leq x)$, standardize both sides of the inequality, and use the CDF of the standard Normal:

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Differentiating that expression, with the chain rule, gives the PDF.

6.4 The 68-95-99.7 rule

Theorem (68-95-99.7 rule). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{P}(|X - \mu| < \sigma) \approx 0.68, \quad \mathbb{P}(|X - \mu| < 2\sigma) \approx 0.95, \quad \mathbb{P}(|X - \mu| < 3\sigma) \approx 0.997.$$

These are three benchmarks for the Normal distribution: the probabilities of falling within one, two and three standard deviations of the mean. It is often easier to apply the rule after standardizing, in which case

$$\mathbb{P}(|Z| < 1) \approx 0.68, \quad \mathbb{P}(|Z| < 2) \approx 0.95, \quad \mathbb{P}(|Z| < 3) \approx 0.997.$$

The rule says that 95 percent of the time, a Normal random variable falls within ± 2 standard deviations of its mean. A more accurate approximation says that 95 percent of the time it falls within ± 1.96 standard deviations. This is why the number 1.96 appears so often in statistics.

7. Summary

	Discrete	Continuous
f is called	the mass function	the density function
$f(x)$ is	$\mathbb{P}(X = x)$	not a probability; may exceed 1
Constraint	$\sum_x f(x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
$\mathbb{P}(X = x)$	$f(x)$	0 at every x
$F(x)$ is	$\mathbb{P}(X \leq x)$	$\mathbb{P}(X \leq x)$
From f to F	sum f up to x	integrate f up to x
From F to f	difference F at x	differentiate F at x
F looks like	jumps and flat regions	a smooth increase
Interval probability	$\sum_{a \in A} f(a)$	$\int_a^b f(x) dx = F(b) - F(a)$

Source: Blitzstein and Hwang, *Introduction to Probability*, 2nd ed., sections 3.2, 3.6, 5.1, 5.2 and 5.4. Free at <https://probabilitybook.net/>.